

MAT 534 HW07

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Problem 1. Let $\varphi: R \rightarrow S$ be a homomorphism of unital commutative rings. Show that if P is a prime ideal in S , then $\varphi^{-1}(P)$ is a prime ideal in R .

Solution. Let $\psi: S \rightarrow S/P$ be the projection homomorphism. Then $\ker \psi = P$, so $\ker(\psi \circ \varphi) = \varphi^{-1}(P)$, and of course $\psi \circ \varphi: R \rightarrow S/P$ is surjective, so $S/P \cong R/\varphi^{-1}(P)$. Since P is prime in S , S/P is an integral domain, hence so is $R/\varphi^{-1}(P)$, implying $\varphi^{-1}(P)$ is a prime ideal of R . $\square$

Problem 2. Let G be a finite subgroup of the group of units $F^\times$ of field F .

- (a) Let d be a divisor of $|G|$. Show that G contains at most d elements whose order divides d .
- (b) Let $|G| = p_1^{e_1} \cdots p_r^{e_r}$ be the prime factorization of $|G|$. Show that for each $i = 1, \dots, r$, there is an element of order $p_i^{e_i}$ in G .
- (c) Show that G contains an element of order $|G|$, and conclude that G is a cyclic group.

Solution. (a) The polynomial $x^d - 1$ has at most d roots because its degree is d ; this is independent of d being a divisor of $|G|$.

(b) Since G is a finite abelian group, it is a direct product of abelian groups $G_1 \times \cdots \times G_r$ where $|G_i| = p_i^{e_i}$. Let $p_i^{k_i}$ be the largest power of p_i that appears as the order of some element of G ; since there are $p_i^{e_i}$ elements (corresponding to the elements of G_i) whose order is a power of p_i , we must have $p_i^{k_i} \geq p_i^{e_i}$ by part (a), so $k_i \geq e_i$ (hence equals e_i), proving the existence of an element with order $p_i^{e_i}$.

(c) G is a finite abelian group, so it is a direct product of cyclic groups; these cyclic groups must be the r groups $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$ by the order information, so $G \cong \mathbb{Z}/p_1^{e_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_r^{e_r}\mathbb{Z} \cong \mathbb{Z}/|G|\mathbb{Z}$ by the Chinese remainder theorem. $\square$

Problem 3. Let $\mathbb{F}_p$ be the field with p elements.

- (a) Let $g(x) \in \mathbb{F}_p[x]$ be an irreducible polynomial with degree $n \geq 1$. Show that $F = \mathbb{F}_p[x]/(g(x))$ is a field with $q = p^n$ elements.
- (b) Show that $g(x)$, when considered as a polynomial in $F[x]$, has a root in F .
- (c) Show that every element of F is a root of the polynomial $x^q - x$.
- (d) Show that, in the polynomial ring $\mathbb{F}_p[x]$, the gcd of the two polynomials $g(x)$ and $x^q - x$ is not 1.
- (e) Conclude that $g(x)$ divides $x^q - x$ in the ring $\mathbb{F}_p[x]$.

Solution. (a) F is a field because $\mathbb{F}_p[x]$ is a PID and irreducible elements of a PID generate maximal ideals.

Furthermore, if α is the image of x under the projection map from $\mathbb{F}_p[x]$ to F , then $1, \alpha, \dots, \alpha^{n-1}$ spans F as a $\mathbb{F}_p$ -vector space, since any polynomial is in the same coset of $(g(x))$ as its remainder upon division by $g(x)$, which has degree at most $n - 1$.

Additionally, $1, \alpha, \dots, \alpha^{n-1}$ are linearly independent over $\mathbb{F}_p$ because linear dependence would imply that α is the root of a polynomial h of degree $n-1$ with coefficients in $\mathbb{F}_p$, which means $h(x) \in \mathbb{F}_p[x]$ is in $(g(x))$; this is a contradiction.

(b) The element $\alpha \in F$ mentioned in (a) is a root of g because $g(x)$ and 0 are in the same coset of $(g(x))$, i.e. the images $g(\alpha)$ and 0 are equal inside F .

(c) The multiplicative group of F is finite with order $q-1$, so by the previous problem it is cyclic of order $q-1$. Thus $a^{q-1} = 1$ if $a \neq 0$, so $a(a^{q-1} - 1) = 0$ for all $a \in F$, i.e. $x^q - x$ has every $a \in F$ as a root.

(d) The projection homomorphism $x \mapsto \alpha$ onto F sends $x^q - x$ to $\alpha^q - \alpha = 0$, so $x^q - x$ lies in the ideal $(g(x))$, hence is divisible by $g(x)$.

(e) uhhhh see part (d)? □

Problem 4. Let F be a field. Prove that the additive group $(F, +)$ is not isomorphic to the multiplicative group $(F^\times, \cdot)$.

Solution. Let $\varphi: F^\times \rightarrow F$ be a homomorphism. Then $\varphi(-1) + \varphi(-1) = \varphi(1) = 0$ implies $\varphi(-1) = \varphi(1) = 0$, so φ is not bijective, hence cannot be an isomorphism. □

Problem 5. Show that the polynomial $(x-1)(x-2)\cdots(x-n)+1$ is irreducible in the ring $\mathbb{Z}[x]$ for all $n \geq 1$ except $n=4$.

Solution. Let $f(x) = (x-1)(x-2)\cdots(x-n)+1$ and suppose $f(x) = p(x)q(x)$ for some $p(x), q(x) \in \mathbb{Z}[x]$ where p and q both have degree less than n . Then $p(k)q(k) = 1$ for all $k = 1, 2, \dots, n$, so $p(k) = q(k) = \pm 1$. Since p and q agree at $n > \max\{\deg p, \deg q\}$ points, p and q are the same polynomial.

Therefore $f(x) = p(x)^2$, so if f is reducible then n must be even and $d := \deg p$ is $n/2$. This implies $p(k) = 1$ has $n/2$ solutions in $\{1, 2, \dots, n\}$ (otherwise either $p(k)-1$ or $p(k)+1$ has more than $n/2$ roots); let $R = \{r_1, \dots, r_d\}$ be the set of these solutions and $S = \{s_1, \dots, s_d\}$ the remaining elements of $\{1, 2, \dots, n\}$ (i.e. the solutions to $p(k) = -1$). Since $f(x)$ is monic, $p(x)$ is monic, so $p(x)$ is uniquely determined as

$$p(x) = (x - r_1)(x - r_2)\cdots(x - r_d) + 1.$$

There exists $s \in S$ such that $r - s \geq d$ for some $r \in R$ because if $1 \in R$ then R contains at most $d-1$ of the numbers $d+1, d+2, \dots, d+d$, hence one of them is in S (and vice versa if $1 \in S$). For such an s , we conclude that

$$2 = |p(s) - 1| = |(s - r_1)(s - r_2)\cdots(s - r_d)| \geq |r - s| \geq d,$$

so if $d > 2$ (hence $n > 4$) then f is irreducible. If $d = 1$ then $|s - r_1| = 1 \neq 2$, so f is still irreducible when $n = 2$. (When $d = 2$, f is reducible: $(x-1)(x-2)(x-3)(x-4)+1 = (x^2-5x+5)^2$.) □

Problem 6. Let F be a field. Let R be the set of polynomials in $F[x]$ whose coefficient of x is equal to 0. Show that R is a subring of $F[x]$, and that R is not a UFD.

Solution. To show that R is a subring of F we verify that

- $1 \in R$ trivially;
- if $f(x), g(x) \in R$ then the coefficient of x in $f(x) + g(x)$ is $0 + 0 = 0$; and
- if $f(x), g(x) \in R$ then the coefficient of x in $f(x)g(x)$ is $c_f \cdot 0 + 0 \cdot c_g = 0$ where c_f, c_g are the constant terms of $f(x), g(x)$ respectively.

However, R is not a UFD because $x^6 = x^2 \cdot x^2 \cdot x^2 = x^3 \cdot x^3$ has two factorizations into irreducibles; both x^2, x^3 are irreducible because any factorization of either one must have a linear factor. $\square$

Problem 7. Show that the ideal $(x, y)^n = (x^n, x^{n-1}y, \dots, y^n)$ in the ring $\mathbb{Q}[x, y]$ cannot be generated by fewer than $n + 1$ elements.

Solution. For $r \in \mathbb{Z}_{\geq 0}$, let I_r be the ideal $(x, y)^r$ and suppose that I_n is generated by the polynomials $p_1(x, y), \dots, p_m(x, y) \in \mathbb{Q}[x, y]$. Since $p_i(x, y) \in I_n$, each monomial of p_i has degree at least n .

The quotient of $\mathbb{Q}[x, y]$ -modules I_n/I_{n+1} is isomorphic as a $\mathbb{Q}$ -vector space to the collection of homogeneous polynomials in $\mathbb{Q}[x, y]$ (since the quotient kills off all monomials with degree at least $n + 1$, and I_n a priori has all monomials appearing with degree at least n), so it has dimension $n + 1$ as a $\mathbb{Q}$ -vector space (one example of a basis is $x^n, \dots, xy^{n-1}, y^n$). On the other hand, an element $p(x, y)$ of I_n can be written as

$$p(x, y) = \sum_{i=1}^m q_i(x, y)p_i(x, y)$$

for $q_1(x, y), \dots, q_m(x, y) \in \mathbb{Q}[x, y]$. When the non-constant terms of $q_i(x, y)$ multiply with $p_i(x, y)$, they produce monomials that have degree at least $n + 1$, so, projecting onto I_n/I_{n+1} ,

$$p(x, y) = \sum_{i=1}^m c_i \cdot p_i(x, y)$$

where c_i is the constant term of $q_i(x, y)$. Thus each element of I_n/I_{n+1} is manifested as a $\mathbb{Q}$ -linear combination of the p_i , implying that the p_i span the $(n + 1)$ -dimensional vector space I_n/I_{n+1} , so $m \geq n + 1$. $\square$

Problem 8. Let R be a UFD, and F its field of fractions.

- Show that if $p \in R$ is irreducible, then the constant polynomial p is irreducible in $R[x]$.
- Show that if $f(x) \in R[x]$ is primitive and irreducible in $F[x]$, then $f(x)$ is irreducible in $R[x]$.
- Suppose that we have two factorizations

$$p_1 \cdots p_k \cdot f_1(x) \cdots f_m(x) = q_1 \cdots q_\ell \cdot g_1(x) \cdots g_n(x)$$

in the ring $R[x]$, with $p_i, q_i \in R$ irreducible and $f_i(x), g_i(x) \in R[x]$ primitive and irreducible in $F[x]$. Prove that $k = \ell$, $m = n$, and that the factors on both sides are equal up to reordering and multiplication by a unit in R .

Solution. (a) If p has a factorization into non-units in $R[x]$, the factors both have degree zero, so the factors would be in R ; p being irreducible in R means such a factorization doesn't exist.

(b) If $f(x)$ has a factorization into non-units, this factorization must be of the form $p(x)q(x)$ for some non-constant $p(x), q(x) \in R[x]$ (we can assume non-constant because f is primitive). However,

this factorization is then also a factorization into non-units in $F[x]$, contradicting the irreducibility of $f(x)$.

(c) By multiplying each polynomial by a unit (and absorbing the unit into one of the p_i, q_i) we can force all the f_i and g_i to be monic, so $p_1 \cdots p_k = q_1 \cdots q_\ell$ and $f_1(x) \cdots f_m(x) = g_1(x) \cdots g_n(x)$. The former equation, together with R being a UFD, implies that the p_i and q_i are the same up to reordering.

For the $f_i(x)$ and $g_i(x)$, we induct on m : specifically, we show that if an element of $R[x]$ is expressed as the product of m irreducible primitive polynomials, then this product is unique. This is trivially true at $m = 1$, so assume that the conclusion holds for all $k \leq m$. Since $F[x]$ is a PID, irreducible elements are prime, so $f_m(x) \mid g_1(x) \cdots g_n(x)$ implies that $f_m(x)$ divides, hence is equal (up to a unit in F) to one of the g_i , say WLOG $g_n(x)$ by relabeling. Since the f_i, g_i are all primitive, this unit in F is actually a unit in R , so we can cancel f_n and g_m to get that $f_1(x) \cdots f_{m-1}(x) = g_1(x) \cdots g_{n-1}(x)$, at which point the inductive hypothesis tells us that the multisets $\{f_1, \dots, f_{m-1}\}$ and $\{g_1, \dots, g_{n-1}\}$ are equal, finishing the proof. $\square$